Mohammad Saif

+91 9170718677 | ms22002200@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

M.Tech Data Science graduate (CGPA: **9.08**) with hands-on experience in **Machine Learning, Generative AI, RAG, LLMs, and Data Engineering**. Built **3+ AI projects**, including a hybrid RAG document assistant, an agentic AI travel planner, and predictive ML models. Achieved **R^2 = 0.984** in groundwater-level prediction and secured **1st place in a hackathon** involving analysis of **50,000+ records**. Proficient in **Python, Scikit-learn, LangChain, Hugging Face, PostgreSQL, Airbyte and dbt**, with experience developing ML solutions and end-to-end data pipelines.

EDUCATION

Jamia Millia Islamia University, Delhi | M.Tech in Data Science — Aug 2024 – Jun 2026
*CGPA: **9.08***

Chaudhary Charan Singh University, Meerut | B.Tech in Information Technology — Sep 2020 – May 2024
*CGPA: **7.52***

TECHNICAL SKILLS

AI / Gen AI: LangChain, Hugging Face, Transformers, LLMs, RAG, Prompt Engineering, AI Agent, Tool Calling, Embeddings
LLM & Retrieval: Groq, Gemini, Ollama, FAISS, BM25, FlashRank, Hybrid Retrieval
Machine Learning: Scikit-learn, TensorFlow/Keras, PyTorch, Regression, Classification, Feature Engineering, Model Evaluation
Data Engineering: Airbyte, dbt, PostgreSQL, SQL, Data Quality
Data Analysis: Pandas, NumPy, Matplotlib, Seaborn, Statistics, EDA
Backend & Tools: Python, Flask, Streamlit, Git, GitHub, Jupyter Notebook, FastAPI

WORK EXPERIENCE

AICTE – Edunet Foundation | AI/ML Intern — Jun 2025 – Jul 2025

- Processed datasets **100k+ records across 10+ domains**, performing data cleaning, EDA, and feature engineering using Python, Pandas, NumPy.
- Implemented and explored **6 ML algorithms**: Linear Regression, Logistic Regression, KNN, Decision Tree, Random Forest, and SVM.
- Developed **2 predictive modeling projects**: Employee Salary Prediction (Regression) and Student Education Loan Prediction (Classification).
- Experimented with **5 KNN neighborhood values** and applied feature scaling, train-test splitting (75:25), and model comparison.
- Evaluated regression and classification models using **5+ evaluation measures**, including MAE, MSE, RMSE, R^2, Accuracy, Precision, Recall.

ZestBotz | Data Engineering Intern — Apr 2025 – Jun 2025

- Integrated OpenWeather API data for **15+ cities**, capturing more than **20 weather attributes** to track temperature, humidity, atmospheric pressure, wind speed, and weather conditions.
- Designed an **Airbyte–PostgreSQL ingestion pipeline**, 8 hours **refresh intervals** to collect, store, and organize historical weather observations.
- Developed **5+ dbt SQL models** for staging, cleaning, aggregation, and city-wise and time-based analytics, targeting **100K+ weather observations**.
- Performed EDA and implemented **5+ data-quality checks** to identify missing values, duplicate records, invalid measurements, and inconsistent timestamps

PROJECTS

TravelPilot | *Personal AI Travel Planning & Trip Optimization Agent* — Jan 2026 – Mar 2026

- Developed a Streamlit agentic AI travel-planning system using **LLM-based** intent understanding and dynamic tool calling to generate personalized multi-day itineraries.
- Built a hybrid architecture combining LLM reasoning with **deterministic Python service**s for budget validation, geographic clustering, route optimization, and schedule feasibility.
- Implemented **multi-factor itinerary** scoring using budget fit, geographic efficiency, travel pace, and weather suitability; integrated PostgreSQL persistent memory and FastAPI REST APIs.

SmartDoc AI | *Advanced RAG Document Assistant* | GitHub — Nov 2025 – Dec 2025

- Built a RAG pipeline supporting **7 document formats**, using **800-character chunks, 150-character overlap**, and MiniLM embeddings.
- Implemented **hybrid search** with RRF to fuse the **Top-25 candidate passages**, reranking them with FlashRank to select the **Top-5 relevant passages**.
- Integrated **FlashRank cross-encoder reranking** to refine retrieved passages before LLM generation, enabling source-grounded answers with document and page-level citations.
- Integrated 4 LLMs for cloud/local LLM inference and developed **6 productivity tools** summarization, quiz, flashcard, and document workflows.

Groundwater Level Prediction & Analysis using MLR and ANN — Aug 2025 – Oct 2025

- Developed an end-to-end prediction pipeline integrating **5 public data sources** covering groundwater, soil, demographic, irrigation, and environmental data for Delhi NCR.
- Processed and analyzed **20+ dataset columns** using data cleaning, missing-value handling, EDA, and cyclical sine/cosine feature engineering.
- Built and compared **MLR and ANN models** using a chronological 70:30 train-test split, achieving **R^2 = 0.984 and RMSE = 1.275** with ANN versus **R^2 = 0.972** with MLR.

ACHIEVEMENTS

- 1st Position – SCRIET Hackathon: Led a team analyzing 50K+ records, deriving insights, and developing a web dashboard for analysis.

CERTIFICATIONS

- Accenture – Digital Skills: Artificial Intelligence | 2026 | Overall Score: 83%. — Certificate
- Udemy – Complete Data Science and Training | 2026. — Certificate